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Hybrid gender agreement pattern in Polish

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INTRODUCTION

Language can be a surprisingly complex system, especially for something used by people every day with such ease. We generate complex structures with words linked to each other in intricate ways all the time and somehow other people manage to understand the meaning we try to convey through this whole system of communication.

Polish is an example of a language with a relatively free word order, therefore often the links between words are marked morphologically. Agreement between words is one of the ways to figure out if certain words are connected, which means the two words share a value of a certain feature they both have. Though it usually is very simple, sometimes agreement can be extremely convoluted. This study aims to investigate at least some of this complexity to help find ways to describe the agreement patterns and universalities between languages.

CHAPTER 1

THEORETICAL BACKGROUND

1.1 Previous work on hybrid agreement

Agreement can be found between a word with some inherent features (a controller) and another, related word (a target). We can see attributive agreement between a demonstrative and a noun in (1), where the noun is plural, so the demonstrative has to be *these* to agree with the number of the noun. In (2) there is predicate agreement: the subject is the singular pronoun *she*, so the verb has to agree with the number and take the form *reads*. These are examples of agreement marking a syntactic dependency, but it can also be used to mark a discourse dependency. This can be found in (3), where the personal pronoun *she* has to be feminine to have the same referent as the earlier noun phrase *little sister*.

- (1) **These dogs** belong to my neighbor.
- (2) **She reads** every morning.
- (3) I have a little sister, **she's** 5 years old.

There are cases where the justification for an agreement is rather semantic than syntactic. This can be found in (4): the copular *are* has to be plural, despite the nouns in the subject noun phrase being singular – this is because there is a group reading of the coordinate phrase and there is plural agreement to show that.

- (4) John and Mary **are** rich.

Hybrid agreement is possible when there are agreement alternatives, but the controller remains the same. Corbett (2023) provides a simple example of that in English: both sentences in (5) are acceptable. The controller of agreement here is the word *family*, the same in both sentences, and there are two agreement alternatives: *have* in (5a) and *has* in (5b). One of them is available, which is known from the lexical semantics of the noun: it can refer to a group of people. The other alternative comes from the morphosyntactic specification: the noun is singular.

- (5) a. The family have lost everything.
- b. The family has lost everything.

This is an example of hybrid number agreement, but same is possible for any other feature that requires agreement in a given language. This study focuses on gender agreement in Polish. Before the system of genders in Polish is described, this section describes Corbett's work on agreement, as it is the main basis for the study conducted later.

Corbett proposed the Agreement Hierarchy, which orders different types of agreement. This hierarchy is used to formulate the constraint visible in (6).

attributive > predicate > relative pronoun > personal pronoun

Figure 1.1: The Agreement Hierarchy (Corbett, 1979)

- (6) *Constraint of the Agreement Hierarchy (to be revised)*

For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with greater semantic justification will increase monotonically (that is, with no intervening decrease).

We can see this on the example of the word *family*. Starting with the leftmost element of the Agreement Hierarchy, (6a) shows that singular (syntactic) agreement is acceptable, but (6b) shows that plural (semantic) is not. However, with the next step of the hierarchy, predicate agreement, both options are possible, as (5) has shown. Plural agreement is also available with a personal pronoun, which can be found in (7).

- a. This family
- b. * These family

- (7) I saw my family yesterday. They seemed to be tired.

This constraint was created for the core instances of hybrid agreement, such as the one shown above (a lexical hybrid), and constructional mismatches, an example of which can be seen in (8).

- (8) The girl and the boy have been on the playground.

Both nouns in the coordination are singular, but the number in the auxiliary verb *have* agrees with the plural number suggested by the whole construction, instead of the number of the nouns within the construction.

The problem with the Agreement Hierarchy constraint in the form shown above is that it is based on “semantic justification”, when hybrid agreement can occur without one of the agreement options having a clear semantic justification. This is clear when we expand the plane of possible hybrid agreements and look at what else can trigger agreement. Alternative sources of agreement information are visible on the left of Table 1.2. The table

also gives an idea of where hybrid agreement comes from: it happens whenever different sources give conflicting information about the value of a given feature. This is represented by the thick line which separates the situations where the source suggests syntactic agreement (above the line) from the situations where it suggests semantic agreement (below the line).

		controller type			
		split hybrids	lexical hybrids	constructional mismatch	extraneous
Hierarchy of Agreement Sources	cell specification (1)				
	morphosyntactic specification (2)				
	lexical semantics (3)				
	complex nominal constructions (4)				
	extraneous source (5)				

Figure 1.2: Core (in gray) and generalized semantic agreement.

This figure includes two more examples of hybrid agreement: the split hybrid *ręka* (meaning 'hand') in Polish and the Norwegian *pancake* sentences with extraneous agreement information.

Polish number system used to have three possible values: singular, dual and plural. Dual is not used anymore, but the word *ręka* (meaning 'hand') is one of few where the relics of dual number can be found: the form used historically as locative dual can now be used as locative singular. In modern Polish the form morphologically resembles masculine or neuter, while the noun itself is feminine. This leads to a situation, where attributive agreement with the locative singular form of the noun can be masculine, as shown in (9).¹

- (9) w lewym ręku
 in left-sg.loc.M hand-sg.loc
 in the left hand

There are two conflicting sources of agreement: the morphosyntactic specification of the noun is feminine, but the specification of the (singular locative cell) can be masculine/neuter. The question then is: which of the options can be considered

¹In this example the M1, M2 and M3 forms are syncretic; whenever that is the case, the M label will be used.

“semantic” agreement? One could say the feminine is semantic, because it makes it possible to track the antecedent later in the discourse, but this explanation can be considered to stretch the notion of “semantic” agreement to cover a less-than-semantic example.

On the other hand there is Norwegian “default agreement”, that is triggered when there is no straightforward agreement controller, for instance when a sentence contains logical metonymy. As an example Corbett uses the sentence visible below.

- (10) Magnus Carlsen og Bobby Fischer hadde vær-t **super-t**.
 Magnus Carlsen and Bobby Fischer had be-pst.ptcp **super-sg.n**
 ‘Magnus Carlsen and Bobby Fischer would have been wonderful.’

Here, by *Magnus Carlsen and Bobby Fischer* speaker means *having Magnus Carlsen and Bobby Fischer play chess*. Instead of agreeing with the features suggested by the coordination, there is extraneous (default) agreement. Here the conflicting sources of agreement are: coordination of proper nouns that refer to two men and the metonymy, which triggers the “default agreement”. Here the example is considered to be more than semantic and therefore not fitting in the core semantic agreement.

Including split hybrids and examples with extraneous agreement information in the analysis forces us to expand beyond semantic agreement and consider agreement options that are less or more than semantic. This is what Corbett did by proposing generalized semantic agreement, along with the revised constraint of Agreement Hierarchy.

- (11) *Revised constraint of the Agreement Hierarchy*

For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with a greater degree of generalized semantic justification will increase monotonically (that is, with no intervening decrease).

The aim of the current paper is to see whether the Polish gender agreement pattern really is compatible with the Agreement Hierarchy and with the Hierarchy of Agreement Sources. Polish has a rich gender system, there is gender agreement with adjectives, some verbs and pronouns and it has large lexical and syntactic resources available. First, it is necessary to understand the system of grammatical gender in Polish.

1.2 Grammatical gender in Polish

There are multiple options of defining grammatical genders in Polish. The one chosen for this study was proposed by Mańczak (1956), where he differentiates between 5 genders: masculine personal, masculine animate, masculine inanimate, feminine and neuter. The difference between the genders is the easiest to observed when looking at appropriate forms of adjectives in the accusative case. Adjectives agree their gender with nouns and the accusative case provides the most diversity in terms of gender forms.

The example presented in Table 1.3 comes from Mańczak (1956). There are 5 nouns and one adjective for all of them used as an example: *mężczyzna* (meaning 'man'), *pies* (meaning 'dog'), *stół* (meaning 'table'), *kobieta* (meaning 'woman'), *dziecko* (meaning 'child') and *dobry* (meaning 'good').

singular		plural		gender of the noun
adjective	noun	adjective	noun	
dobrego	mężczyznę	dobrych	mężczyzn	masculine personal
	psa		psy	masculine animate
dobry	stół		stoły	masculine inanimate
dobrą	kobietę		kobiety	feminine
dobre	dziecko		dzieci	neuter

Figure 1.3: Examples of nouns in accusative with 5 grammatical genders in Polish with their agreeing adjectives.

The table shows that each of the nouns agrees with a different set of adjective forms in the accusative singular and plural: *-ego* and *-ych* form for masculine personal, *-ego* and *-e* for masculine animate, *-y* and *-e* for masculine inanimate, *-ą* and *-e* for feminine and *-e* in both numbers for neuter. There are 5 possible sets of agreements, which gives 5 possible grammatical genders.

Saloni (1976) proposed an even more elaborate system, where he talked about 9 genders (instead of one neuter there were two and additionally there were three genders for plurale tantum nouns). Such a system is, however, not necessary for the current analysis, therefore for the sake of simplicity we will adopt the 5 genders proposed by Mańczak. For ease of reference we will use for them the names that Saloni used, which are: M1 (masculine personal), M2 (masculine animate), M3 (masculine inanimate), F (feminine) and N (neuter).

1.3 Polish nouns with possible hybrid gender agreement

Corbett (2023) provides the example of a split hybrid *ręka* (meaning 'hand') in Polish, though unfortunately that example cannot be explored further with the resources available for this study. The circumstances under which the noun has developed to its current use are rare and there are no other nouns which can be analyzed in addition to this one. There are, however, three other categories of nouns which can be used to study hybrid gender agreement and these are described in the following subsections. This work focuses agreement with nouns and does not address complex nominal constructions, such as coordination.

1.3.1 Socially gendered nouns

Here the socially gendered nouns are defined as the nouns that refer to people and semantically involve the gender or sex of the referent. Corbett (2023) provides the German *das Mädchen* (meaning 'girl') as an example of lexical hybrids – the grammatical gender of the noun is neuter, but the semantics of the word include the feminine gender. The nouns in this group are similar to the German example. There is a number of words in Polish where the gender of the referent is incongruent with the grammatical gender of the noun, which can lead to hybrid agreement. An example, very similar to the German one, can be found in (12).

- (12) To dziewczę jest niemożliwe! Ona w ogóle się nie uczy!
 this-N girl-N is impossible-N she in general REFL not study
 'This girl is impossible! She never studies!

Here the lexical hybrid is essentially the same as the German *das Mädchen* – a noun meaning 'girl' has neuter grammatical gender. In the example we find three words agreeing with the noun: demonstrative *to* agreeing with the grammatical gender, predicate adjective *niemożliwe* agreeing with the grammatical gender and the personal pronoun *ona* in the second sentence agreeing with the semantic gender.

The socially gendered nouns are an instance of lexical hybrids: the conflicting sources of agreement are between the lexical semantics and the morphosyntactic specification of the noun. Most of those nouns refer primarily to women or girls in an expressive (often derogative) manner, but there are different combinations of conflicting values of gender. The process of choosing the nouns for this category and the combinations of conflicting genders are described in the Methodology chapter.

1.3.2 Incongruent profession nouns

In Polish some nouns referring to people holding certain job titles and functions or practising certain professions have masculine gender and there are no feminine nouns with the same meanings. These nouns are likely to be understood to only refer to men, but for a lack of better term they can be also used to refer to women. This can be seen in the examples (13) and (14).

- (13) Nowa psychiatra przepisała mi nowe leki.
 new-F psychiatrist prescribed-F me new medicine
 (14) Nowy psychiatra przepisał mi nowe leki.
 new-M psychiatrist prescribed-M me new medicine

The same noun *psychiatra* is used in both sentences and the only way to tell whether the referent is female or not is by looking at the agreement: feminine in (13) and masculine in (14).

Another option to mark the referent as female is to precede the masculine noun by the word *pani*, which can be translated as 'lady' or 'madam'. It is however possible to omit it.

With recent initiatives to ensure that women are linguistically represented in all fields, there is more discussion about those masculine nouns, specifically for which of them it is difficult to create the feminine counterpart and why is it so (Szpyra-Kozłowska, 2019; Szpyra-Kozłowska & Laidler, 2022). The current study takes advantage of this area of linguistic research by using some examples of those masculine nouns as opportunities to find hybrid gender agreement.

1.3.3 Depreciative forms of M1 nouns

In his paper Saloni points out that for all M1 nouns in plural nominative and vocative case it is possible to create an alternative form, which he calls "depreciative" (Saloni, 1988). This form is created very systematically, specifically by inflecting the noun the same way a noun of a different gender would be inflected. The M1 noun *profesor* (meaning 'professor') will serve as an example here. *Profesor* represents the nominative singular form of the noun. There are two options for the nominative plural form of this noun: *profesorzy* and *profesorowie*, both of these are non-depreciative. To create the depreciative form, it is necessary to look for a noun of a different gender with the same ending. For this example we can use the M3 noun *procesor* (meaning 'processor'), which in plural nominative takes the form *procesory*. Therefore the depreciative form of the noun *profesor* is *profesory*.² The example (15) shows the depreciative form used in a sentence, where both attributive agreements and the predicative agreement is non-M1, therefore agreeing with the gender in the specification of the cell.

- (15) Te stare profesory przyszły w zabłoconych
 these-M2/M3/N/F old-M2/M3/N/F professor-M1 came-M2/M3/N/F in muddy
 butach.
 shoes.
 These old professors came in with muddy shoes.

Despite the name, the purpose of using a depreciative form is not necessarily to deprecate the referent, although it has been found to be a common function of the form. Based on a corpus study, Siuda (2010) reports that the purpose of 66% depreciative forms is to express negative feelings towards the referent, 24% of the forms are used as a stylistic device, 6% are supposed to reduce the distance to the referent and 4% have a different function.

Depreciative forms can be created only for M1 nouns, but they are created by inflecting them as if they were of any other gender. The conflicting sources of agreement are the following: the morphosyntactic specification of the nominative (and vocative) plural is M2, M3, N or F, while the morphosyntactic specification of the noun is M1. This makes depreciatives the only form used in this study that represents a different controller type: a split hybrid.

²In plural nominative and vocative forms are syncretic, therefore the same goes for the vocative form.

1.4 Hypothesis

The aim of this study is to investigate the extent to which Corbett's constraints of the agreement hierarchy hold on Polish data with hybrid gender agreement. The hypothesis is the following: Polish examples of hybrid gender agreement (socially gendered nouns, profession nouns and depreciative nouns) fit the simple constraint of the agreement hierarchy: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement for all kinds of nouns tested. More semantic agreement is expected in predicate agreement than in attributive and even more in relative pronoun agreement. Furthermore, according to the Hierarchy of Agreement Sources, there should be more semantic agreement with the lexical hybrids (so the socially gendered and profession nouns) than with the split hybrids (depreciative forms).

CHAPTER 2

METHODOLOGY

2.1 Agreement sources

The corpus used for this study was the 300M subcorpus of the National Corpus of Polish (Przepiórkowski, 2012), which was parsed using Trankit (Nguyen, Lai, Veyseh, & Nguyen, 2021). The dependency trees (conforming to the Universal Dependency standard, henceforth UD) obtained this way were searched for words, which could trigger hybrid agreement: profession nouns, socially gendered nouns and depreciative forms of M1 nouns. The dependencies of those words then made it possible to search for attributive agreements (adjectives of nouns), predicate agreements (verbs agreeing with their subjects) and relative pronoun agreement. A tagged corpus could have been sufficient for attributive and predicate agreement, but using a parsed corpus gave a better chance of finding the correct relations and agreements within them and made it possible to include relative pronouns in the analysis. The subsections below describe how the nouns for analysis were chosen.

2.1.1 Profession nouns

Nouns in this category came from two studies about feminitives in Polish. The first was a survey reported by Szpyra-Kozłowska (2019), where participants were asked to create feminitives from 50 masculine nouns, that in modern Polish do not have feminine counterparts at all or those counterparts are rarely used. For each noun participants had the option to give no answer. The second study was an experiment conducted by Szpyra-Kozłowska and Laidler (2022), where they tested whether certain feminitives are less accessible by asking participants to try to pronounce them. They chose 20 feminitives, which are formed from masculine nouns that have consonant clusters such as /kt/ (e.g. *architekt*, meaning 'architect'), /tr/ (e.g. *geriatra*, meaning 'geriatrician') or /rg/ (e.g. *chirurg*, meaning 'surgeon'). According to Klemensiewicz (1957) adding the -ka suffix, the function of which can be to create the feminitive, makes the word difficult to pronounce. Those two studies give a list of 60 nouns in total, for which it is said to be

difficult to form feminine counterparts. It is therefore likely that, if there was a female referent of a noun from this list, the masculine noun would be used, leading to hybrid agreement.

Both of the studies determine in some way which of the tested nouns have less available feminines than others: in Szpyra-Kozłowska and Laidler (2022) those are the nouns for which there are the highest error rates in pronunciation, whereas in Szpyra-Kozłowska (2019) – the nouns for which the most people refused to propose a feminine. Taking this approach has, however, two downsides: first, it requires an arbitrary threshold that separates the nouns that have an easily formed feminine from the nouns for which creating the feminine counterpart is more difficult. Second, this approach significantly limits the number of nouns for analysis. Considering this, the current study used all of the nouns from the two studies on Polish feminines.

For ease of reference, all of those nouns are called *profession* nouns, despite not all of them actually referring to professions. Meanings of all nouns are presented in Table 4.1 in the Appendix.

2.1.2 Socially gendered nouns

The nouns for this category were found using the Polish Academy of Sciences Great Dictionary of Polish available online.¹ The advanced search option allowed to search for nouns in the thematic category HUMAN AS A PHYSICAL BEING: HUMAN PHYSICALITY: SEX. This search yielded 586 results, which then had to be manually filtered to find the nouns that only refer to humans (and not, for example, abstract nouns or nouns referring to body parts) and that had incongruent grammatical and semantic gender. There were 19 nouns selected this way: 7 of them were grammatically masculine animate and referred to women (e.g. *babsztyl*, meaning 'woman' in a pejorative way), 1 was grammatically masculine inanimate and referred to women (*pasztet*, referring to a woman whom the speaker finds ugly or unattractive), 7 were grammatically neuter and referred to women (e.g. *pannisko*, meaning 'a young woman' in an expressive way), 2 were grammatically neuter and referred to men (e.g. *chłopaczysko*, meaning 'a tall or big young man') and 2 were grammatically feminine and referred to men (e.g. *ciota*, meaning 'a homosexual man' in a pejorative way). Those nouns can be found in Table 4.1 in the Appendix.

2.1.3 Depreciative forms

For this category no list of nouns was created, since the corpus was annotated and it was possible to look up the depreciative forms by checking the words' morphological description. It is possible to create depreciative forms of surnames, however these were excluded from the analysis, because there was no way of determining the referents and therefore no way of ensuring that there is hybrid agreement. Surname *Machala* found in the corpus can serve as an example: let us say that the *Machala* family bought a new car.

¹<https://wsjp.pl/>

Assuming the family consists of at least one man, the speaker can use both (16a) and (16b) to convey that information, depending on their attitude they have towards the situation: (16a) if the speaker is neutral or positive and (16b) if they are negative. However, if there are only women in the family, the speaker can only use (16b). Without the context it is unclear whether *Machały* in (16b) is used as a depreciative or simply to refer to women and therefore we cannot decide if this is an example of hybrid agreement.

- (16) a. Machał-owie kupi-li nowe auto.
 Machała-pl.nom.m1.ndepr buy-3pl.m1.past new car
 b. Machał-y kupi-ły nowe auto.
 Machała-pl.nom.m1.depr buy-3pl.f.past new car
 “Machałas bought a new car.”

Morphological analyzer Morfeusz (Kieraś & Woliński, 2017) can mark forms as surnames which made it easy to exclude them from the analysis.

2.1.4 Finding agreement sources in the corpus

The code used for extracting corpus data in this study can be found in a GitHub repository.² The general idea of how the code works is presented in this chapter.

After choosing the nouns to be searched in the corpus, it was necessary to find all of their forms and the forms’ morphological descriptions, which was done using the morphological analyzer Morfeusz. First the corpus was searched for occurrences of nouns, which could be sources of hybrid agreement. Those nouns were saved if they were in the list of gendered or profession noun forms or if they were marked as depreciative in their morphological description. To ensure that no homonyms were accidentally saved for later analysis, the morphological description of a found word was compared with the one given by Morfeusz. This way, for example, the word *premiera* was excluded if it was an instance of the lexeme *premiera*, meaning *premiere* but not if it was an instance of the lexeme *premier*, meaning *prime minister*, in the singular accusative form.

Once it was sure that the right noun was found, the dependency tree was searched for words that can agree their gender with that of the noun. This was the head of the noun (which could be an example of predicate agreement), adjectives attached to the noun (which could be examples of attributive agreement), and relative pronouns referring to the noun. Based on the UD annotation guidelines, all dependents of the word were searched for a dependency label with a subtype :RELCL, which in the UD annotation scheme mark the head of the relative clause. The relative pronoun was attached to the head of the relative clause and had a UD feature PRONTYPE=REL.³

Each occurrence was additionally annotated for the distance between the source and potential target of agreement (measured in intervening words) and the ordering: whether

²<https://github.com/bmagdab/hybrid-agreement>

³https://universaldependencies.org/workgroups/newdoc/relative_clauses.html

the target of agreement was the first or not. Once the possible agreement targets were found, the data could be used to look for occurrences of hybrid agreement.

2.2 Listing agreement occurrences

To analyse hybrid gender agreement in the found examples, it is necessary to mark the alternative to the grammatical gender of each agreement source. For all profession nouns that alternative gender was feminine, because the only chance of hybrid agreement for those noun is when the gender of the referent is feminine. For socially gendered nouns the alternative gender was assigned depending on the lexeme.

Once the alternative gender was determined, it was necessary to exclude plural nouns for which the gender discrepancy was between feminine and neuter. This is because for all types of agreement tested here, the plural feminine and plural neuter forms of targets of agreement are syncretic, therefore there is no hybrid agreement in those cases. This happens, for instance, in (17), where in singular ((17a) and (17b)) there is a difference between the feminine and neuter forms of the word *młoda*. In plural (17c), however, the form is the same for both neuter and feminine, so hybrid agreement cannot be observed. Examples like this were therefore excluded.

- (17) a. *młod-e* *dziewczę*
 young-sg.n.nom girl.sg.n.nom
- b. *młod-a* *dziewczę*
 young-sg.f.nom girl.sg.n.nom
- c. *młod-e* *dziewczę-ta*
 young-pl.nom girl-pl.n.nom
 “young girl”

The next step was to find occurrences of different types of agreement. The predicate agreement of a given noun was found in its head on three conditions: first, dependency label between the noun and the head had to be NSUBJ, which is the UD label for nominal subjects. Second, the head had to be a verb that inflects for gender, which is only true for verbs annotated as past participles,⁴ active and passive participles. Third, the noun has to be in the nominative case. This condition helps filter out situations where a numeral phrase is in subject position, because then the verb does not agree with the subject – it is in 3rd person, singular and neuter form, regardless of the noun. Therefore, predicate agreement cannot be observed in those situations at all and they should be excluded.

The search was simpler with attributive and relative pronoun agreement. The only requirement for attributive agreement was that the word is an adjective, which was verified by looking for the tag ADJ in the morphological description of the dependent. As for relative pronouns, all of the ones found earlier were included in the analysis, unless

⁴The annotation scheme used for NKJP uses the past participle label not only for past participles, but also for parts of past tense and conditional mood forms (Przepiórkowski, 2012).

they did not inflect for gender (which was the case, for example, for the word *gdzie*, meaning *where*).

A sentence from the 1 million subcorpus of the National Corpus of Polish with a dependency tree in (2.1) serves as an example of the process of extracting agreement described in this chapter. The word, which could trigger hybrid agreement (but here does not, since the referent is male) is *premierem*. There are two words in the sentence that agree with this word in terms of gender: *jedynym* and *który*. All of the information saved for each instance of agreement is shown in Table 2.2.

		Occurrence 1	Occurrence 2
type of text		typ_qmow	
source of agreement	form	premierem	
	lexeme	premier	
	morphological description	subst:sg:inst:m1	
	grammatical gender	m1	
	other gender	f	
agreement target	form	jedynym	który
	morphological description	adj:sg:inst:m1:pos	adj:sg:nom:m1:pos
	gender	m1	m1
	distance	0	0
	target first	1	0
	semantic agreement	0	0
type of agreement		attributive	relative pronoun

Figure 2.2: All of the information extracted from the sentence in (2.1)

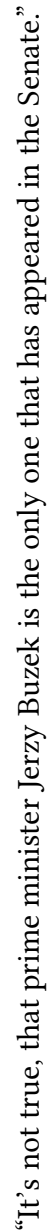


Figure 2.1: Dependency tree for the example sentence.

CHAPTER 3

RESULTS

3.1 Descriptive statistics

The first hypothesis this study aims to test is: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement. Additionally, the hypothesis proposed with the definition of generalized semantic agreement will be tested, which is that split hybrids (here the depreciative forms) are less likely to have generalized semantic agreement than lexical hybrids (here the socially gendered nouns and profession nouns). Before those hypotheses are addressed, let us look at the dataset extracted from the corpus.

A total of 199 927 observations of agreement have been collected, although a vast majority of them was agreement with profession nouns, for which it was impossible to clearly determine whether it was hybrid agreement or not, as this would require manually finding the referent of the noun from the context. Because of that, only profession nouns with feminine agreement were used in the analysis. The summary of the types of agreements and controllers found in the dataset is in Tables 3.1 and 3.2 respectively.

First, we will look at proportions of semantic agreement with attributive, predicate and relative pronoun targets, which are visualized by Figure 3.1. Semantic agreement was favored in predicate agreement with both depreciatives and socially gendered nouns. This is contrary to the hypothesis about monotonous shift from one value to another along the agreement hierarchy, but this will be further addressed in the next section.

Type of agreement	N	Percentage
attributive	1873	52.627%
predicative	1478	41.529%
relative pronoun	208	5.844%

Table 3.1: Numbers and percentages of different types of agreement in the dataset.

Type of controller	N	Percentage
depreciative	1040	29.222%
gendered	807	22.675%
profession	1712	48.103%

Table 3.2: Numbers and percentages of different types of controllers in the dataset.

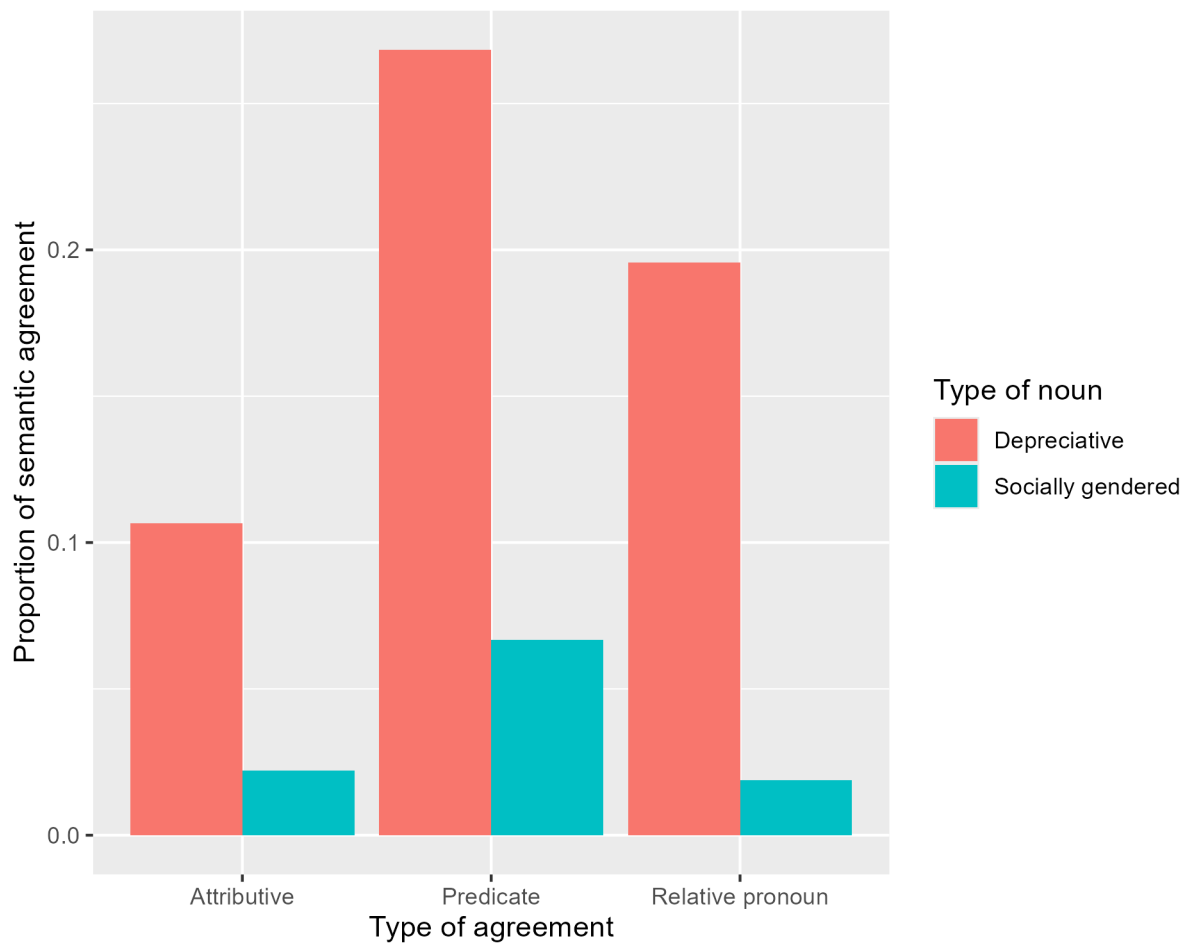


Figure 3.1: Proportions of semantic agreement for each type of agreement for socially gendered and depreciative nouns.

Type of agreement	N	Percentage
Attributive	845	49.357%
Predicate	758	44.276%
Relative pronoun	109	6.367%

Table 3.3: Number of different types of agreement found with profession nouns along with their percentages among the semantic agreements with profession nouns

Type of agreement	Type of noun	Average distance
attributive	depreciative	0.3223350
	gendered	0.3801262
predicative	depreciative	1.6233333
	gendered	1.9416667
relpron	depreciative	0.5652174
	gendered	0.6981132

Table 3.4: Average distance between the target and the controller in all types of nouns and agreements as shown in Figure 3.2.

For profession nouns only feminine agreement is considered, which means all of the examples have semantic agreement, so these occurrences cannot be reported in a similar figure. It can be reported, however, how many of the semantic agreement occurrences with profession nouns were attributive, predicate and relative pronoun agreement. This can be found in Table 3.3. We can see a similar number of attributive and predicate, but very few relative pronoun agreements.

Figure 3.2 shows the average distance between the target and the controller of agreement measured as the number of intervening words. It was the largest in predicate agreement for both depreciatives and socially gendered nouns. The exact average distances can be found in Table 3.4. It was most common for attributive agreements to have the target before the controller (73.5%), with predicate agreements it happened in 30.2% of cases. There were 11 occurrences of target before controller in relative pronoun agreement, but those have been a result of parsing errors since it is not possible to put a relative pronoun before the noun to which it refers.

As for the hypothesis regarding the hierarchy of agreement sources, 20.4% of agreements with depreciatives had semantic agreement, while it was only 2.85% for socially gendered nouns. This is further looked into in the following section.

It should be noted there were 15 profession nouns that did not have semantic agreement at all. Those nouns, along with their translations to English, can be found in Table 3.5.

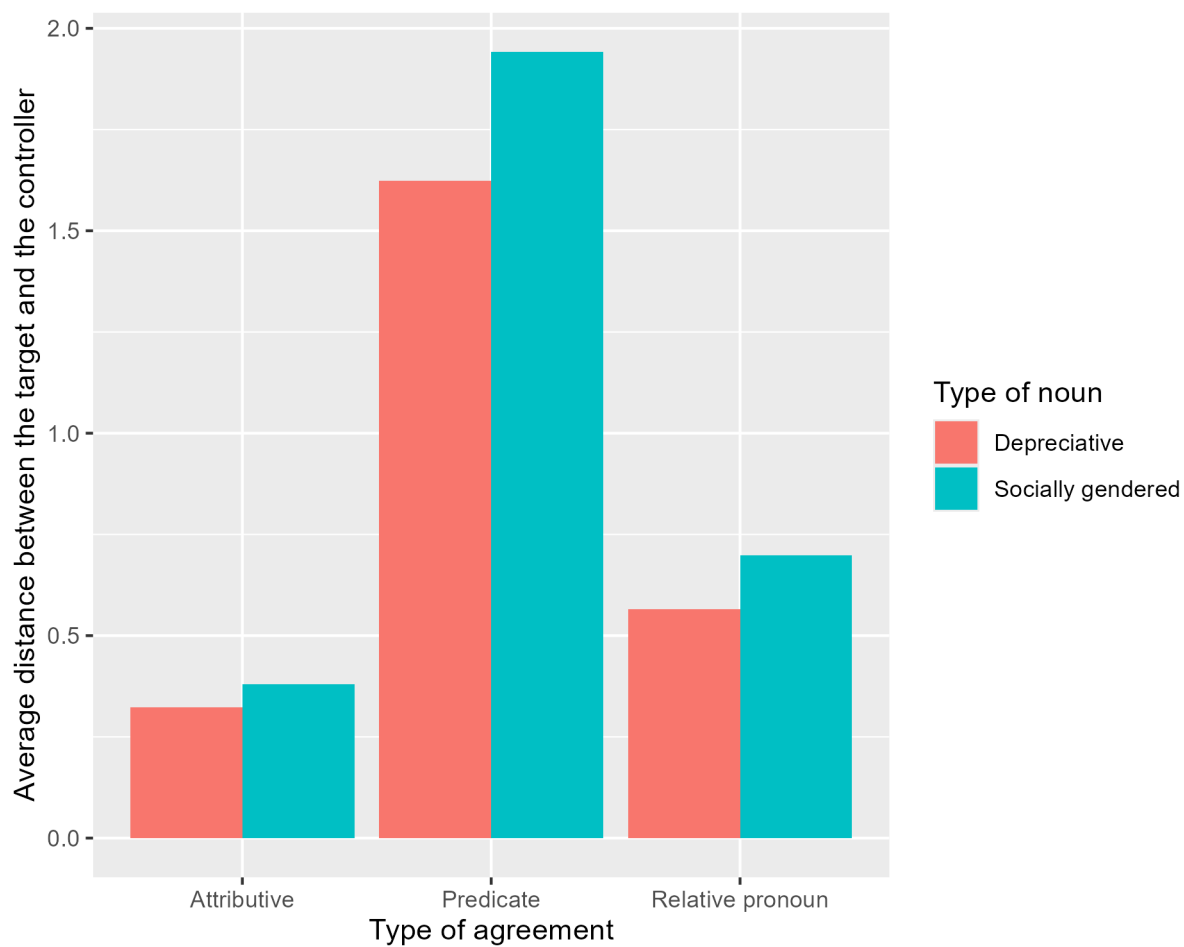


Figure 3.2: Average distance between the target and the controller of agreement for each type of agreement and for each type of noun.

Lexeme in Polish	Translation to English	No. of occurrences
demiurg	demiurge	119
dramaturg	playwright	397
elekt	person elected in a specified post, but has not yet entered office	14
foniatra	phoniatrician	2
ftyzjatra	pulmonologist	1
geometra	geometer	39
geriatria	geriatrician	3
jaskiniowiec	caveman	35
metalurg	metallurgist	8
murarz	bricklayer	241
muzykolog	musicologist	80
nieuk	dunce	37
nurek	diver	109
pajac	clown	10
szklarz	glazier	50

Table 3.5: Table of profession nouns (and their translations and numbers of occurrences in the corpus), for which there was no semantic agreement found in the dataset.

3.2 Statistical models

The models discussed in this section are not based on the profession nouns, since a vast amount of data in this category contained occurrences of the nouns that could not have hybrid agreement and could not have been filtered out automatically. The following analysis is based only on the agreements with depreciative forms and socially gendered nouns.

The analysis was conducted using the `glm` function from the `lme4` R package. Models computed using this function were all binomial generalized linear models, because the dependent variable was binary: whether semantic agreement was chosen or not. There were 4 variables tested as predictors for semantic agreement in separate generalized linear models. Those variables were: type of agreement (attributive, predicate, relative pronoun), distance between the target and the controller, target preceding the controller and type of controller noun (socially gendered or depreciative). No random factors were included. The formulae used for the models as well as the full output can be found in the Appendix.

The model predicting based on the type of agreement showed that predicates favor semantic agreement more than attributives (estimate = -1.6722, $p < 2e-16$), but relative pronouns favor semantic agreement less than predicates (estimate = 1.0042, $p = 0.0036$), which does not show a monotonous shift from one value of gender to the other, as was expected. The model based on the distance gave expected results on the other hand, the larger the distance between the target and the controller, the more likely it is the semantic agreement will be chosen (estimate = 0.14115, $p = 2.32e-06$). According to the model predicting based on the ordering of the target and the controller, target being before the controller lowers the likelihood of semantic agreement (estimate = -1.53658, $p < 2e-16$). Finally, the socially gendered nouns are less likely to have semantic agreement than the depreciatives (estimate = -2.1664, $p < 2e-16$).

Additionally, a multivariable model using the same 4 variables as predictors was computed. Estimates and p-values from this model are presented in Table 3.6, the formula and full output of the model can be found in the Appendix. According to this model, when considering the interaction between all the predictors, only some of them are statistically significant. Predicate agreement favors semantic agreement, while socially gendered nouns do not. Target appearing before the controller also discourages from using semantic agreement.

Predictor	Estimate	P-value
Predicate agreement	0.45490	0.0345
Relative pronoun agreement	-0.21101	0.5959
Distance	0.03983	0.2191
Ordering	-0.99116	1.59e-06
Type of noun: gendered	-1.73335	3.77e-13

Table 3.6: Estimates and p-values assigned to all predictors in the multivariable model. Predictors, that are statistically significant at the 0.05 level are bolded.

CHAPTER 4

DISCUSSION

One of the surprising outcomes of the study is that there is no monotonous change from one value of gender to the other when going from attributive to relative pronoun agreement, which is contrary to one of the hypotheses. This may be a result of using low quality data for analysis – the corpus, from which agreement was extracted, was first parsed automatically and the dependency trees produced this way have not been reviewed. It is possible that some of the words are annotated incorrectly or that there are non-existent dependencies found in the sentences.

It is also worth pointing out that there were very few instances of relative pronoun agreement – 208 compared to 1478 and 1873 instances of predicate and attributive agreement respectively. In the univariable model predicting by type of agreement the difference between the relative pronoun and the predicate agreement is less significant than the difference between the predicate and attributive agreement, while the relative pronoun predictor in the multivariable model is not significant altogether. The unexpected decline in the proportion of semantic agreement between predicate and relative pronouns can be a result of limited relative pronoun data considered in the analysis. One of the possible directions for future research is to focus on hybrid agreement with relative and personal pronouns, since the current study offered little insight into those agreement targets.

There is also a potentially interesting observation to be made about the large mean distance between the target and controller of agreement, specifically in predicate agreement. The distance was found to be a significant predictor for semantic agreement (the larger the distance the more likely it is that semantic agreement will be chosen), but only by the univariable model. The multivariable model shows, that the effect of distance stops being statistically significant when other predictors are taken into account: one of the significant ones is predicative agreement. As a future perspective, it would be beneficial to test the predicative agreement with small distances between targets and controllers, to see if the preference for semantic agreement was here caused by large distance or by predicate targets.

Corbett (1979) observed the ordering of the target and controller of agreement to have an influence over the choice between semantic and syntactic agreement – this was found in

the current study as well. According to both models, univariable and multivariable, if the target of agreement is placed before the controller it is unlikely that the agreement will be semantic. The effect is highly significant for both models.

Finally, another hypothesis tested here was that split hybrids are less likely to have semantic agreement than lexical hybrids. Depreciatives, which are an example of split hybrids, had semantic agreement in over 20% of all occurrences, while the socially gendered nouns had semantic agreement in only 2.85%, which is contrary to the expectations. It could be interesting to compare the depreciative forms with different split hybrids, to see if this preference for semantic agreement is exclusive to depreciatives or not. In an experimental study it would be possible to test the agreement with the Polish *ręka* (meaning 'hand') used by Corbett (2023) to introduce split hybrids, since the examples in the corpus were not sufficient to do so in the present study.

The corpus analysis yielded many occurrences of agreements with profession nouns, but unfortunately many of those were not useful for the present study. Without manually selecting the occurrences it was impossible to differentiate the examples where the referent was male from the ones where the referent was female, but all agreements were masculine. In the latter case it would have been an instance of hybrid agreement, where the speaker simply chose to use syntactic agreement for every target – an interesting example that is technically possible, but could not have been selected automatically from the whole database of profession noun agreement. Nevertheless, the socially gendered nouns provided the opportunity to study Polish lexical hybrids, so the study was not harmed.

What should be noted, is that 15 of the profession nouns did not have any feminine agreements at all. Most of those words appeared in the corpus less than a 100 times each, the median of their number of occurrences in the corpus was 37. Since those nouns are rare, a better option would be to study them experimentally. There are no obvious femininives for them (Szpyra-Kozłowska, 2019; Szpyra-Kozłowska & Laidler, 2022), therefore there is a high chance of finding hybrid agreement with them.

CONCLUSION

The results of the study show, that with current evidence Corbett's Agreement Hierarchy and Hierarchy of Agreement Sources are not fully compatible with Polish data. The relative pronoun agreement was expected to have more semantic agreement than the predicates, but that was not the case in the analyzed data. This should be further investigated with experimental data, as very few examples of relative pronouns have been found in the corpus. According to the Hierarchy of Agreement Sources there should be more semantic agreement with lexical hybrids rather than with the split hybrids and the study found it to be the opposite. Additionally, profession nouns did not provide as much information as they potentially could have, so further annotation is needed to use them for future analysis.

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APPENDICES

Table 4.1: Socially gendered and profession nouns in Polish and their translations or definitions in English

Noun in Polish	Translation or definition in English	Category
adiunkt	someone who holds a PhD title and is employed at a scientific institution	profession
architekt	architect	profession
babochłop	a woman that resembles a man with her physicality, looks and behavior	gendered
babol	a woman that looks or behaves in a way that causes aversion in the speaker	gendered
babon	a woman, about whom the speaker feels negatively	gendered
babsko	a woman, about whom the speaker feels negatively	gendered
babsztyl	a woman, about whom the speaker feels negatively	gendered
babus	a woman, about whom the speaker feels negatively	gendered
burmistrz	mayor	profession
chirurg	surgeon	profession
chłopaczysko	a tall or big young man	gendered
chłopobaba	a man that resembles a woman with his physicality, looks and behavior (can also mean a woman that resembles a man)	gendered
ciota	a homosexual man (can also refer to a woman about whom the speaker feels negatively)	gendered
demiurg	demiurge	profession
dramaturg	playwright	profession
drań	scoundrel	profession
dupek	asshole	profession
dureń	fool	profession
dziekan	dean	profession

Continued on next page

Table 4.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
dziewczątko	a girl, whom the speaker likes or to whom they feel compassion	gendered
dziewczę	a young girl	gendered
dziewuszyisko	a girl who causes aversion or contempt in the speaker	gendered
ekspert	expert	profession
elekt	person elected in a specified post, but has not yet entered office	profession
foniatra	phoniatician	profession
ftyzjatra	pulmonologist	profession
geometra	geometer	profession
geriatra	geriatrician	profession
ginekolog	gynecologist	profession
głupek	fool	profession
herszt	ringleader	profession
inżynier	engineer	profession
jaskiniowiec	caveman	profession
jeniec	captive	profession
jeździec	rider	profession
kierowca	driver	profession
kobiecisko	expressively about a woman, augmentative, or to refer to a big woman	gendered
kobieciątko	expressively about a woman, diminutive	gendered
kobieton	a woman that resembles a man with her physicality, looks and behavior	gendered
ksiądz	priest	profession
lump	tramp	profession
magister	master (holder of a university degree)	profession
majster	foreman	profession
metalurg	metallurgist	profession
minister	minister	profession
murarz	bricklayer	profession

Continued on next page

Table 4.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
muzykolog	musicologist	profession
mędrzec	wise man	profession
naukowiec	scientist	profession
nieuk	dunce	profession
nurek	diver	profession
pacholę	a boy	gendered
pajac	clown	profession
pannisko	expressively about a young woman, augmentative	gendered
pastor	pastor	profession
pasztet	an ugly and unattractive woman	gendered
pediatra	pediatrician	profession
petent	applicant	profession
polityk	politician	profession
porucznik	lieutenant	profession
powstaniec	insurgent	profession
prefekt	prefect	profession
premier	prime minister	profession
prezydent	president	profession
proboszcz	parish priest	profession
psychiatra	psychiatrist	profession
pułkownik	colonel	profession
rektor	person in charge of a university	profession
sierżant	sargeant	profession
starosta	highest administrative officer of a second-level unit of local government	profession
subiekt	salesman	profession
szklarz	glazier	profession
szpieg	spy	profession
szubrawiec	rascal	profession
upiór	phantom	profession

Continued on next page

Table 4.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
wamp	a confident, misterious, provocative and attractive woman who seduces men	gendered
widz	audience member	profession
wójt	highest administrative officer of a basic-level unit of local government	profession
zbieg	fugitive	profession

```

Call:
glm(formula = semantic_agr ~ type_of_agreement,
     family = binomial(link = "logit"), data = data_for_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -2.0740     0.1238  -16.758   <2e-16 ***
type_of_agreement1 -1.6722     0.1631  -10.249   <2e-16 ***
type_of_agreement2  1.0042     0.3449   2.911    0.0036 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1407.8  on 1846  degrees of freedom
Residual deviance: 1284.3  on 1844  degrees of freedom
AIC: 1290.3

Number of Fisher Scoring iterations: 5

```

Figure 4.1: Output of the model based on type of agreement.


```

Call:
glm(formula = semantic_agr ~ distance, family = binomial(link = "logit"),
     data = data_for_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.07801     0.07966 -26.087  < 2e-16 ***
distance      0.14115     0.02988   4.723 2.32e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1407.8  on 1846  degrees of freedom
Residual deviance: 1386.5  on 1845  degrees of freedom
AIC: 1390.5

Number of Fisher Scoring iterations: 4

```

Figure 4.2: Output of the model based on distance.

```

Call:
glm(formula = semantic_agr ~ target_first, family = binomial(link = "logit"),
     data = data_for_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.30833     0.08401 -15.574  <2e-16 ***
target_first -1.53658     0.16216  -9.476  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1407.8  on 1846  degrees of freedom
Residual deviance: 1301.9  on 1845  degrees of freedom
AIC: 1305.9

Number of Fisher Scoring iterations: 5

```

Figure 4.3: Output of the model based on the order of the target and the controller.

```
Call:
glm(formula = semantic_agr ~ source_cat, family = binomial(link = "logit"),
    data = data_for_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.36243    0.07697  -17.700   <2e-16 ***
source_catgendered -2.16649    0.22512   -9.624   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1407.8  on 1846  degrees of freedom
Residual deviance: 1260.8  on 1845  degrees of freedom
AIC: 1264.8

Number of Fisher Scoring iterations: 6
```

Figure 4.4: Output of the model based on type of controller.

```
Call:
glm(formula = semantic_agr ~ type_of_agreement + distance + target_first +
    source_cat, family = binomial(link = "logit"), data = data_for_model)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.35979    0.21481   -6.330 2.45e-10 ***
type_of_agreementpredicative  0.45490    0.21518    2.114  0.0345 *
type_of_agreementrelpron   -0.21101    0.39788   -0.530  0.5959
distance         0.03983    0.03242    1.229  0.2191
target_first    -0.99116    0.20649   -4.800 1.59e-06 ***
source_catgendered   -1.73335    0.23864   -7.263 3.77e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1407.8  on 1846  degrees of freedom
Residual deviance: 1187.5  on 1841  degrees of freedom
AIC: 1199.5

Number of Fisher Scoring iterations: 6
```

Figure 4.5: Output of the multivariable model.